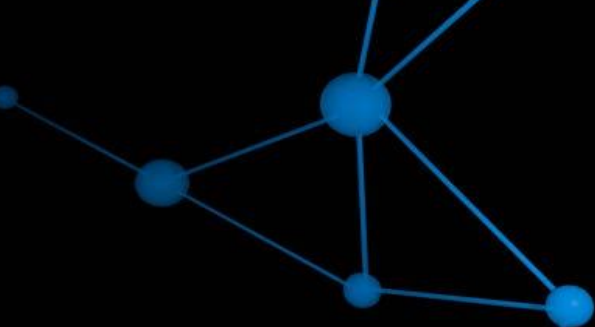
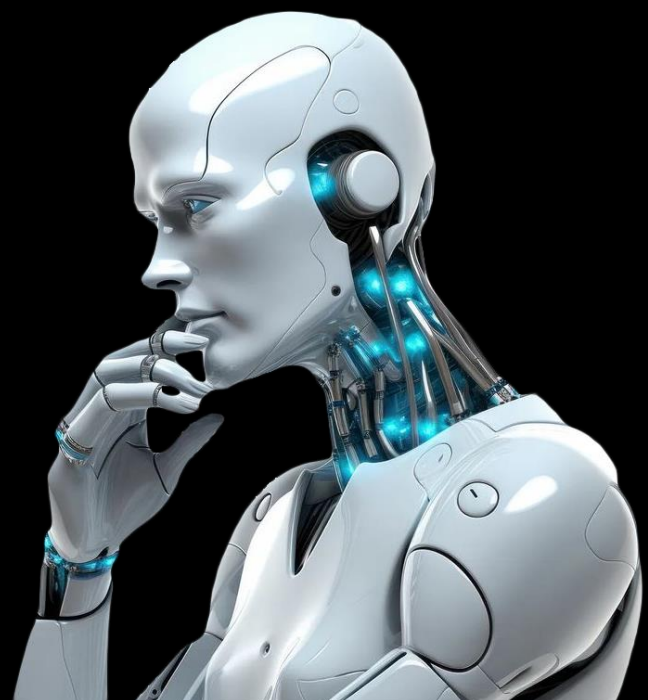




Intelli**Insight**



IntelliInsight

AI-Powered Multi-Agent Enterprise Knowledge & Live Business Intelligence System

Abstract

IntelliInsight is a multi-agent AI-powered enterprise intelligence platform designed to integrate structured data (Excel, SQL databases, Google Sheets links) and unstructured data (PDFs, emails, documents) into a unified intelligent system.

The system dynamically routes data using agent-based orchestration powered by LangChain and LangGraph.

Unlike traditional BI tools such as Microsoft Power BI and Tableau, IntelliInsight combines:

- AI reasoning
- Financial analytics
- Cross-document synthesis
- Real-time spreadsheet monitoring
- Intelligent recommendations

into one enterprise-ready system.



Introduction

Modern organizations generate and store large volumes of heterogeneous data originating from multiple operational and strategic activities. This data can be broadly categorized into **structured** and **unstructured** formats. Structured data includes,

- Financial spreadsheets
- Relational databases
- Google Sheets
- Accounting records
- Key performance indicator (KPI) tables

, that are typically organized in rows and columns. Such data is essential for quantitative analysis, financial forecasting, and operational monitoring.

In contrast, unstructured data consists of documents such as,

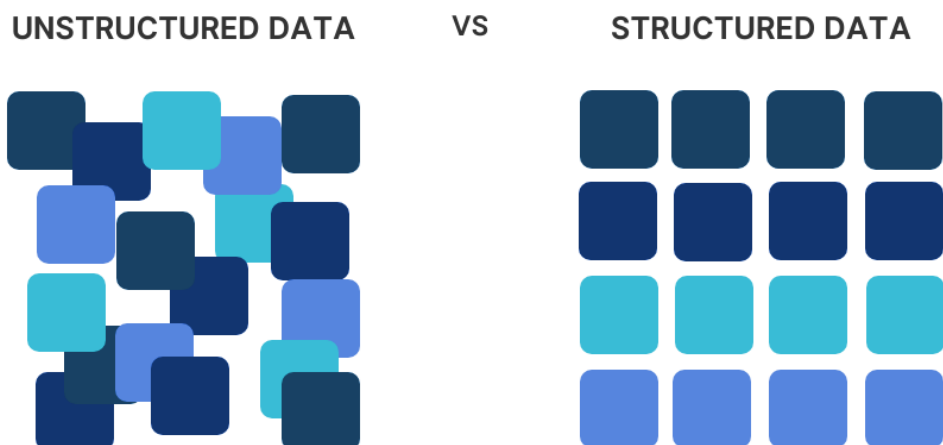
- PDF contracts
- Email communications
- Word reports
- Policy documents
- Meeting minutes

Unlike structured datasets, unstructured data does not follow a predefined schema and often contains contextual and descriptive information that is critical for understanding business decisions, contractual obligations, and organizational processes.

Traditional Business Intelligence (BI) platforms primarily focus on structured data analytics, enabling organizations to generate dashboards, visualizations, and statistical summaries. However, these systems lack the capability to interpret textual documents or provide semantic understanding of unstructured information. On the other hand, **Retrieval-Augmented Generation (RAG)**–based artificial intelligence systems are designed to analyze and answer questions from textual documents, but they typically lack advanced numerical reasoning, financial analytics, and real-time business monitoring capabilities.

As a result, there is currently no unified AI-driven enterprise platform that seamlessly integrates structured and unstructured data sources while simultaneously providing automated financial analysis, intelligent question answering, cross-source insight synthesis, and real-time interactive dashboards. Organizations often rely on separate systems for document analysis and business analytics, leading to fragmented workflows and delayed decision-making.

To address this gap, the proposed system, **IntelliInsight**, introduces a **multi-agent AI-powered enterprise intelligence platform** capable of ingesting **cloud-based spreadsheet links, uploaded financial files, databases, and textual documents**. The system performs **structured financial reasoning, unstructured document understanding, and cross-source synthesis while also providing intelligent answers to user queries**. Through real-time data monitoring, automated anomaly detection, and interactive visualizations, IntelliInsight enables organizations to transform raw enterprise data into actionable insights and informed decision support.



Problem Statement

Organizations face:

- **Fragmented data systems**
- **Manual financial analysis**
- **Delayed business insights**
- **Lack of automated anomaly detection**
- **No AI-based reasoning for decision-making**

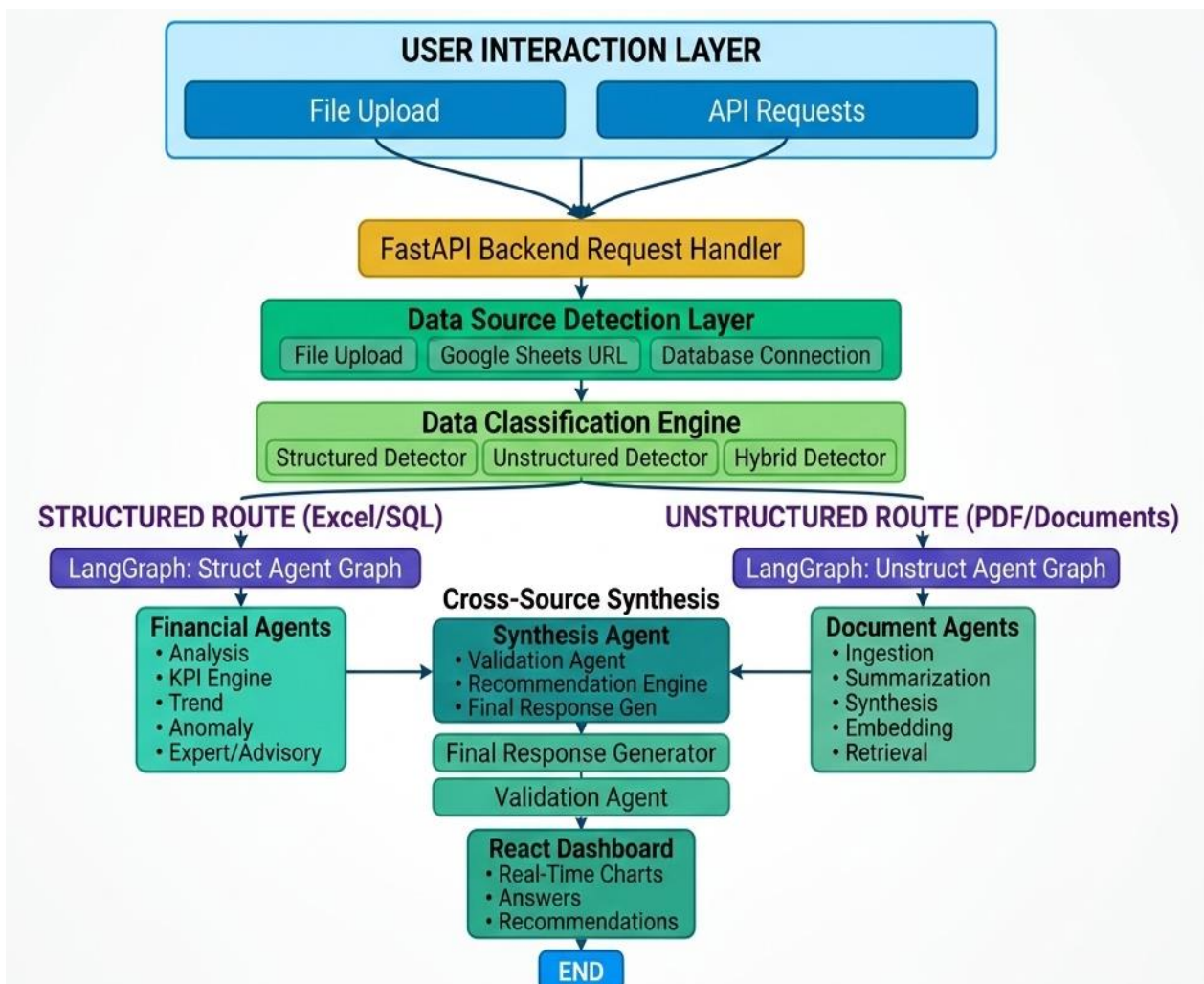


There is a need for:

A unified multi-agent enterprise intelligence system that integrates live structured and unstructured data sources and delivers real-time actionable insights

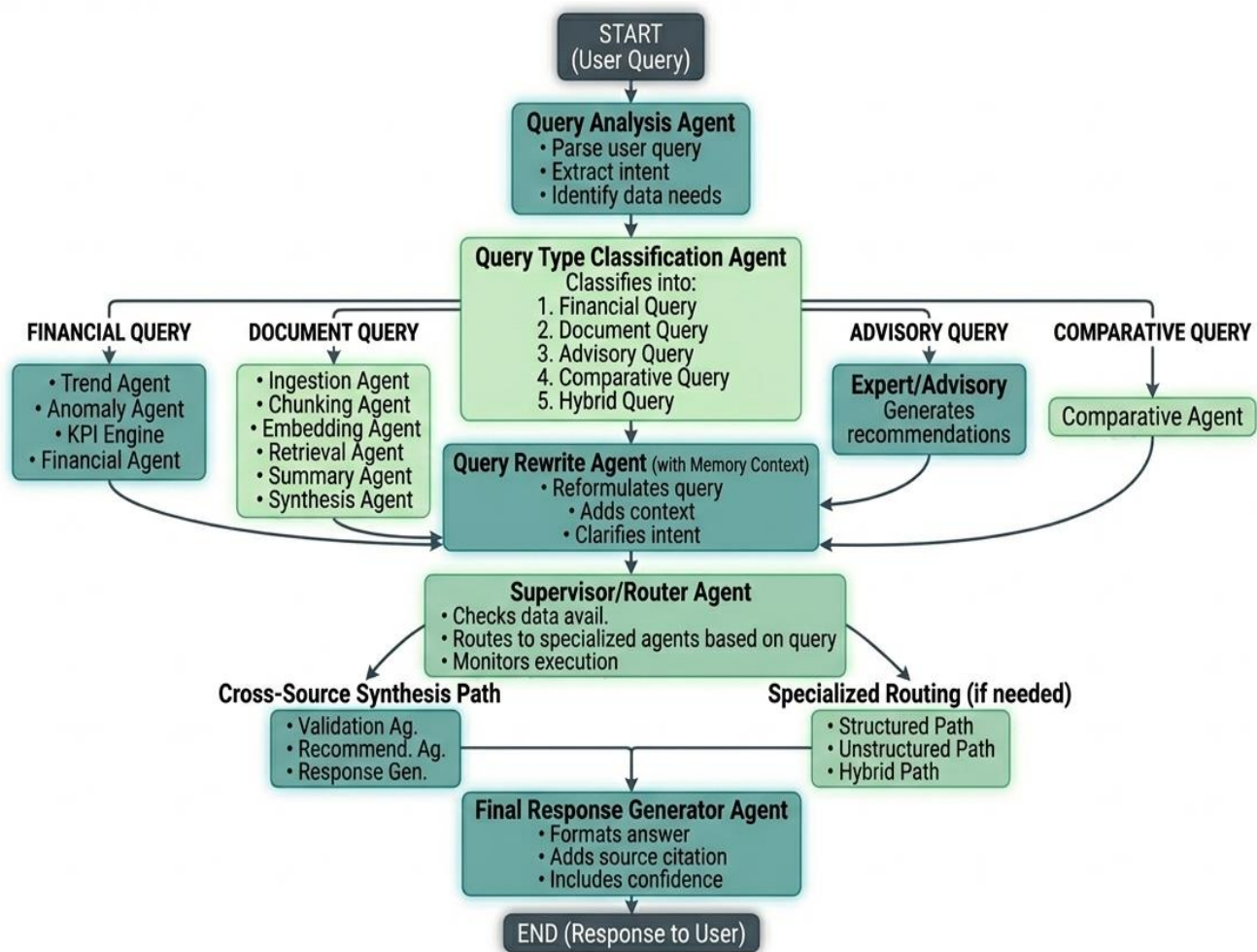
System Architecture Overview

High-Level Architecture



Multi-Agent Architecture

The system uses graph-based orchestration via LangGraph.



MULTI-AGENT SYSTEM SPECIFICATIONS

Agent Categories

Query Processing Agents

- **Query analysis agent** - Parse and understand user query
- **Query Type Classifier** - Classify query type
- **Query Rewriter** - Reformulate with context
- **Supervisor/Router** - Route to appropriate agents

Structured Data Agents (Financial)

- **Trend agent** - Detect trends over time
- **Anomaly Agent** - Identify unusual values
- **KPI Engine** - Calculate key metrics
- **Financial Analysis Agent** - Deep financial analysis
- **Expert/Advisory Agent** - Generate recommendations

Unstructured Data Agents (Documents)

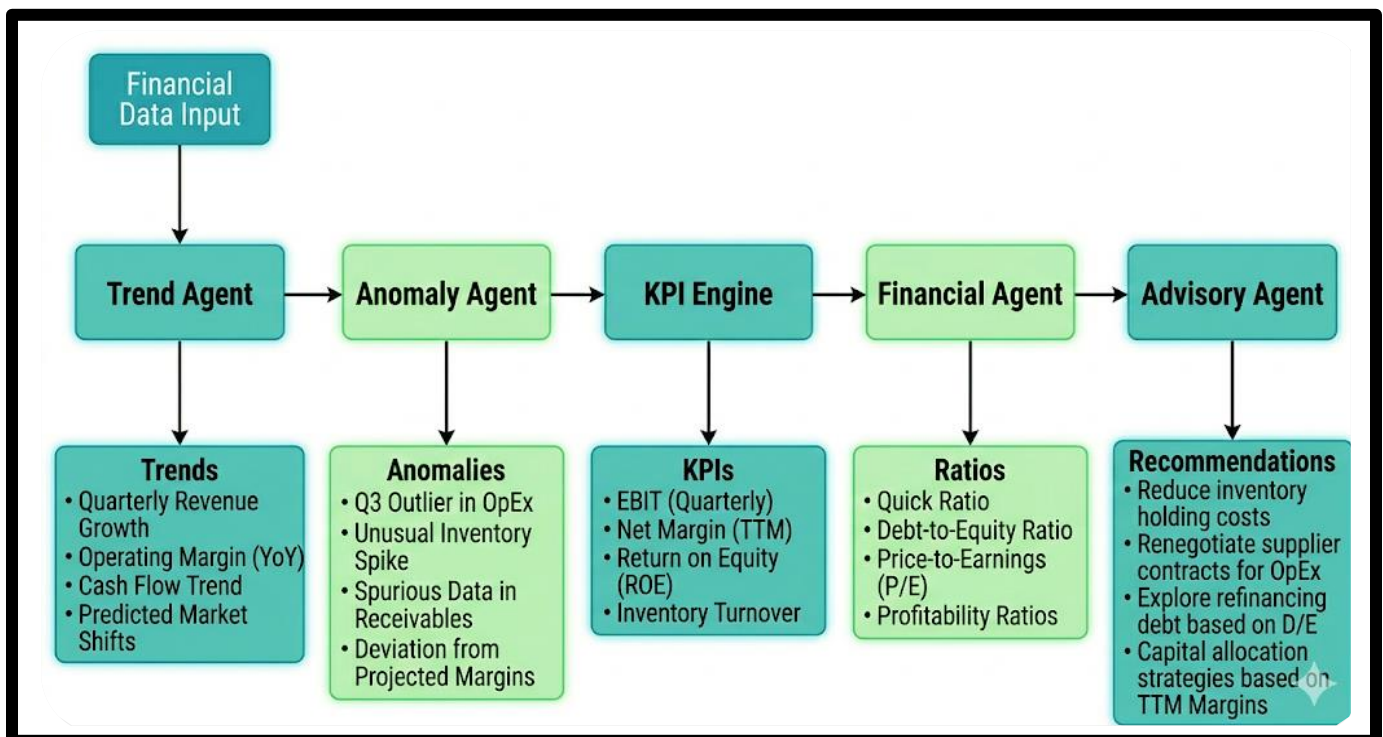
- **Ingestion Agent** - Extract text from PDFs
- **Chunking Agent** - Split into analyzable pieces
- **Embedding Agent** - Convert to vectors
- **Retrieval Agent** - Semantic search
- **Summary Agent** - Create concise summary
- **Synthesis Agent** - Extract key insights

Cross-Source Agents

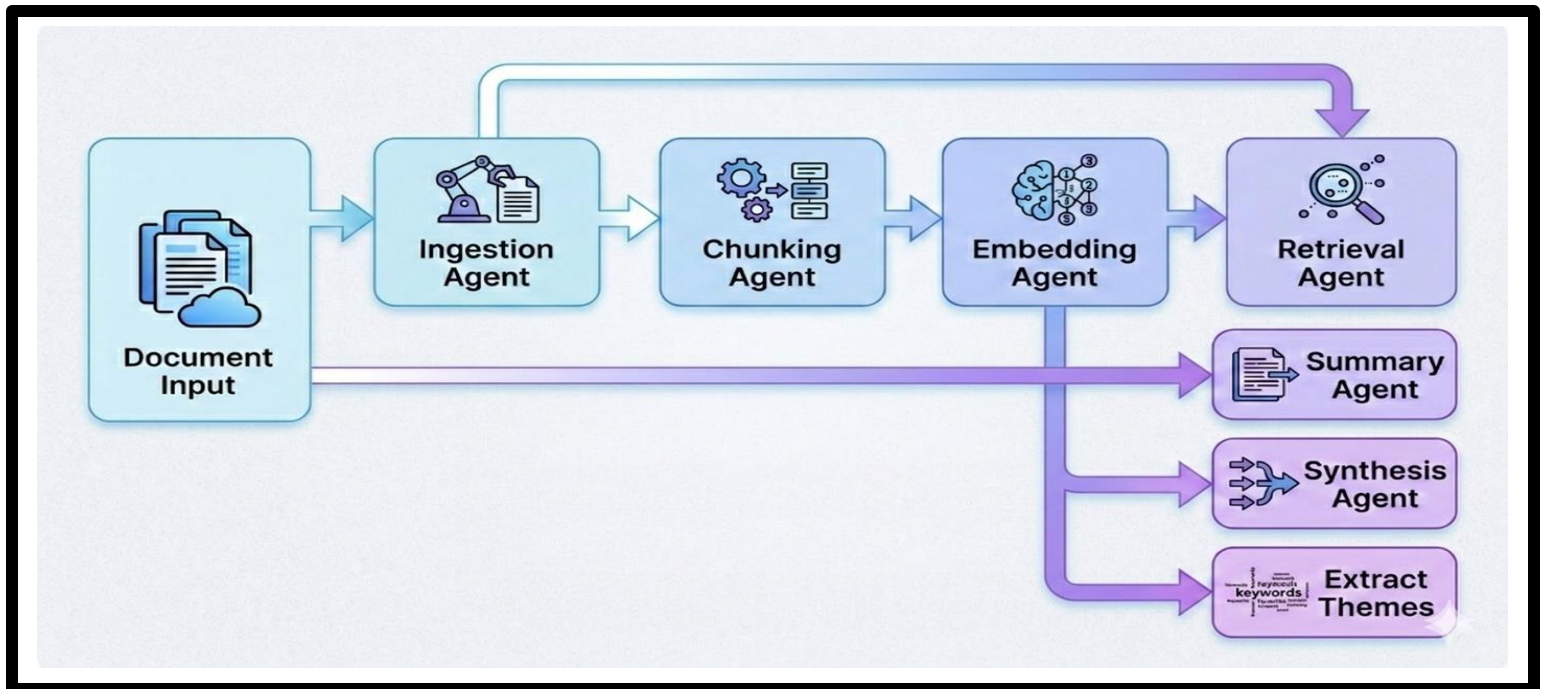
- **Validation Agent** - Verify consistency
- **Recommendation Engine** - Generate insights
- **Final Response Generator** - Format final answer

Agent Interaction Patterns

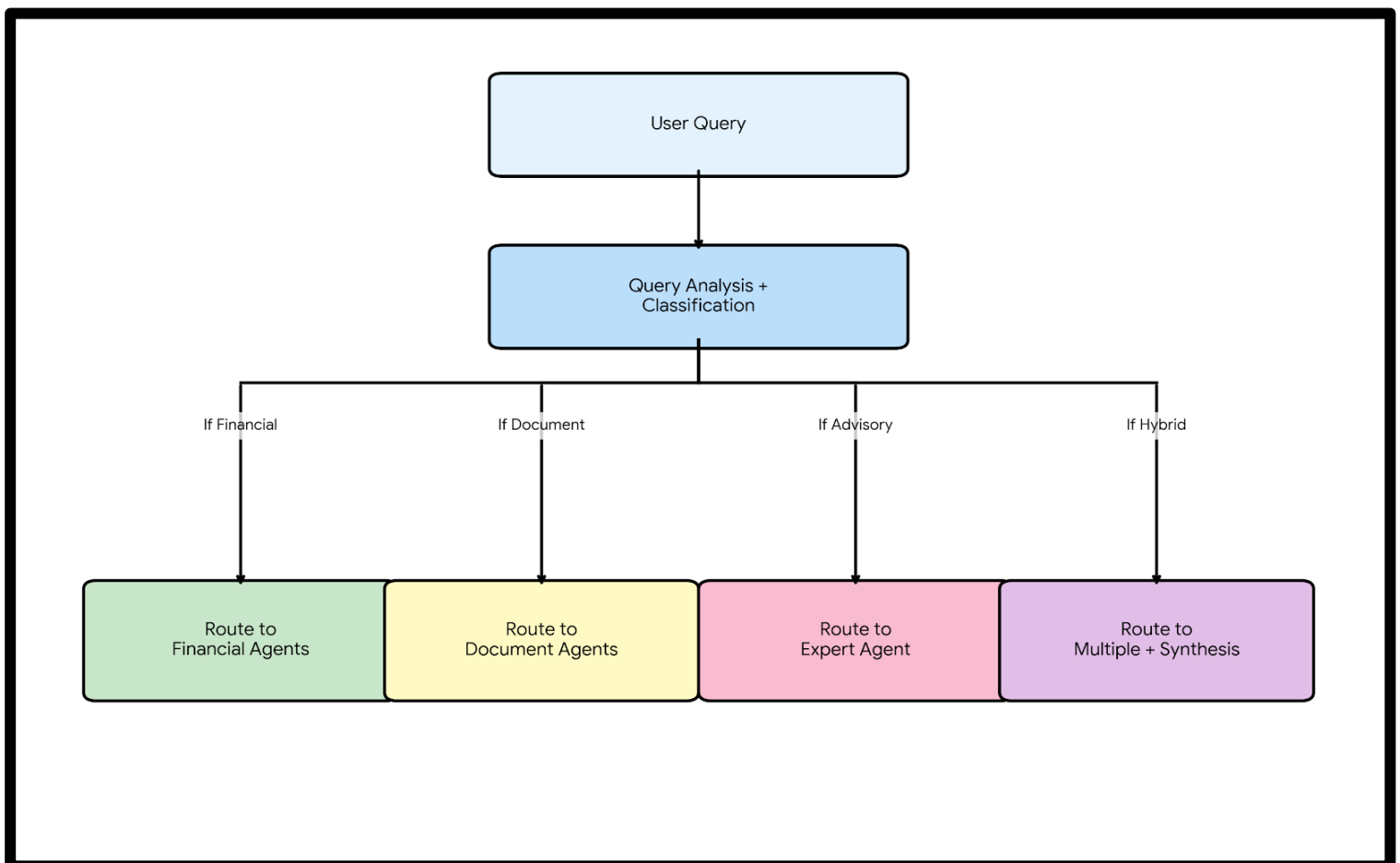
Pattern 1: Sequential Execution (Structured Data)



Pattern 2: Parallel Execution (Unstructured Data)



Pattern 3: Query Routing (Hybrid)



CORE FEATURES

Structured Data Processing (Financial)

Capabilities:

- Real-time Excel/CSV parsing and monitoring
- Google Sheets live data synchronization
- SQL database direct integration
- Automated KPI calculation
- Trend analysis (MoM, YoY, custom periods)
- Anomaly detection with AI explanation
- Financial ratio analysis
- Forecasting and predictions
- Real-time dashboard updates

Example Flow:

User uploads: sales_data.xlsx



Structured Route Detection



Financial Agents Execute:

1. Extract columns and data types
2. Calculate trends (month-over-month)
3. Detect anomalies (unusual spikes/dips)
4. Calculate KPIs (revenue, profit, margin)
5. Generate insights (Why is profit down?)



Real-Time Dashboard

- Sales Trend Chart
- Profit Analysis
- KPI Cards
- Anomaly Highlights

Unstructured Data Processing (Documents)

Capabilities:

- PDF text extraction
- Multi-document ingestion
- Intelligent chunking
- Semantic search (RAG)
- AI summarization
- Insight synthesis
- Q&A over documents
- Cross-document correlation

Example Flow:

User uploads: quarterly_report.pdf



Unstructured Route Detection



Document Agents Execute (Parallel):

1. Extract text and metadata
2. Chunk into analyzable segments
3. Generate embeddings
4. Create executive summary
5. Extract key insights



Document Viewer + Q&A

- Interactive Summary
- Key Themes
- Searchable Chunks
- Answer Questions

Cross-Source Intelligence

Capabilities:

- Correlate structured and unstructured insights
- Answer hybrid queries (EX:- "Why did sales drop? Check emails for context")
- Generate unified recommendations
- Create integrated reports
- Identify related patterns across sources

Example Flow:

User Query: "Why did profit decrease? What should we do?"

Analysis:

Financial Data:

- Profit down 15% (Anomaly Agent)
- Revenue stable, Costs up 20% (Trend Agent)

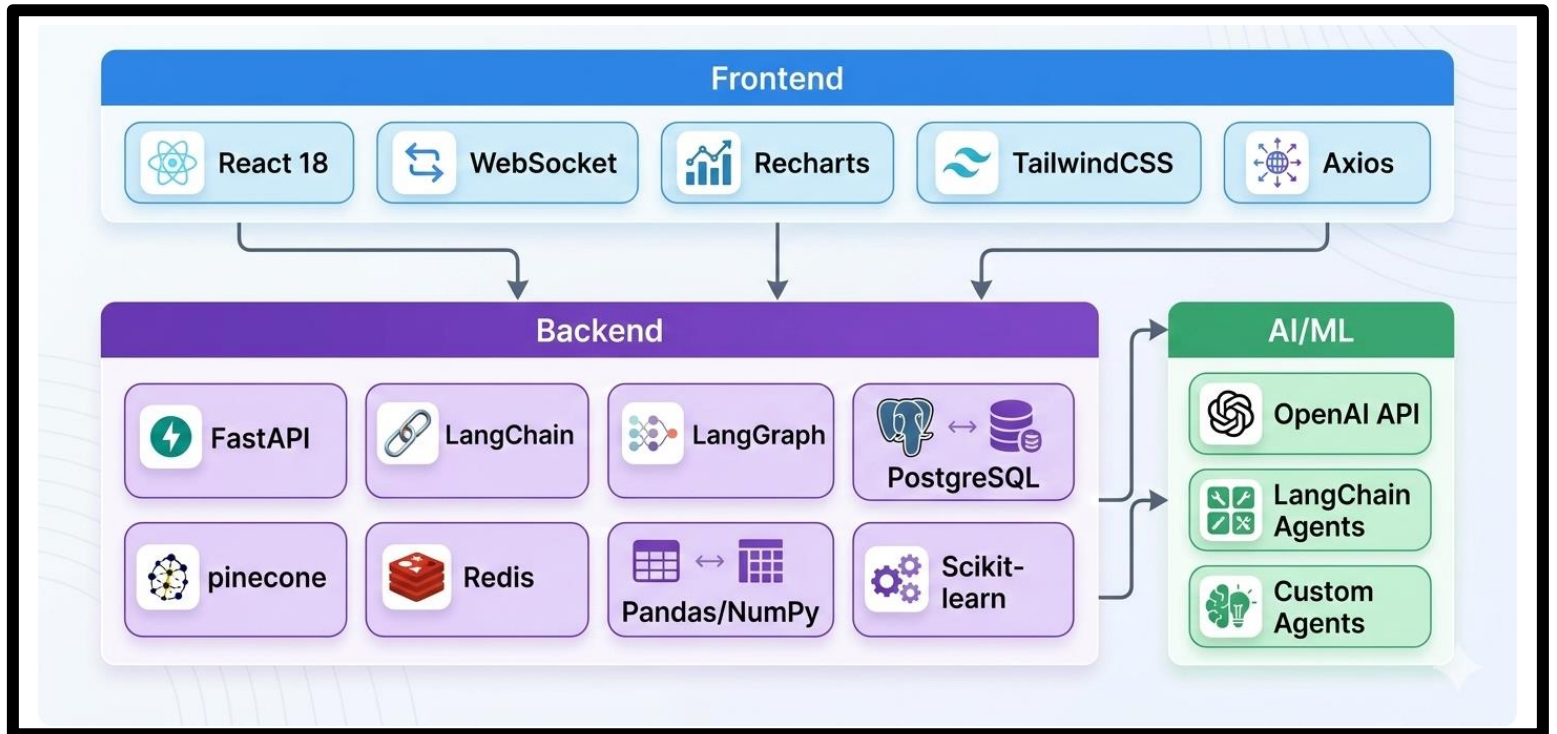
Document Analysis:

- Found email about supplier price increase (Retrieval)
- Contract mentions 5% price hike (Synthesis)

Combined Answer:

"Profit decreased 15% due to a 20% rise in COGS.
Your new supplier contract includes a 5% price increase
effective this month. Recommendation: Negotiate better terms
or find alternative suppliers."

TECHNICAL ARCHITECTURE



Implementation Roadmap

The development of the **IntelliInsight** platform will be executed in seven structured phases over a 12-week timeline. Each phase builds incrementally toward a fully functional, enterprise-grade AI intelligence system.

Phase 1: Foundation (week 1-2)

This phase establishes the core technical infrastructure of the system.

Objectives:

- Set up the FastAPI backend environment
- Initialize the React frontend application
- Configure PostgreSQL database
- Configure FAISS vector database
- Implement base agent class architecture
- Set up LangGraph orchestration environment

Outcome:

A working foundational architecture ready for agent and pipeline development.

Phase 2: Query Processing Agents (week 3-4)

This phase focuses on intelligent query understanding and routing.

Components Developed:

- Query Analysis Agent
- Query Type Classifier
- Query Rewriter Agent
- Supervisor / Router Agent

Outcome:

System can analyze user queries, classify intent (structured/unstructured/hybrid), rewrite for clarity, and route to appropriate pipelines.

Phase 3: Structured Data Pipeline (week 5-8)

This phase enables business intelligence over structured enterprise data.

Components Developed:

- Data Source Detection Module
- Trend Analysis Agent
- Anomaly Detection Agent
- KPI Engine
- Financial Analysis Agent
- Advisory Agent

Outcome:

System can perform financial reasoning, KPI computation, trend analysis, and generate structured business insights.

Phase 4: Unstructured Data Pipeline (week 5-8)

This phase implements document intelligence using RAG architecture.

Components Developed:

- Ingestion Agent (PDF extraction)
- Chunking Agent

- Embedding Agent
- Retrieval Agent
- Summary Agent
- Synthesis Agent

Outcome:

System can process documents, generate embeddings, retrieve relevant chunks, and produce contextual AI-generated answers.

Phase 5: Cross-Source Intelligence (week 5-8)

This phase integrates structured and unstructured outputs into unified intelligence.

Components Developed:

- Validation Agent
- Recommendation Engine
- Final Response Generator
- End-to-end integration testing

Outcome:

System delivers cross-source synthesized insights with validation and actionable recommendations.

Phase 6: Frontend & Real-Time Features (week 8-10)

This phase enhances user interaction and visualization.

Components Developed:

- Interactive dashboard implementation
- WebSocket-based real-time streaming
- Dynamic chart rendering
- Intelligent Q&A interface

Outcome:

Users receive real-time responses, visual dashboards, and an interactive AI-driven experience.

Phase 7: Testing & Deployment (week 10 -11)

This final phase ensures system reliability and production readiness.

Activities:

- Unit testing of agents
- Integration testing of full pipelines
- Performance optimization
- Production deployment

Outcome:

A scalable, tested, and deployable enterprise AI intelligence platform.

CONCLUSION

IntelliInsight represents a significant advancement in enterprise intelligence by:

1. **Unifying** structured and unstructured data processing
2. **Automating** financial analysis and anomaly detection
3. **Enabling** AI-powered reasoning and recommendations
4. **Providing** real-time insights and decision support
5. **Reducing** manual analysis time by 80%+

The multi-agent architecture powered by LangGraph provides:

- **Scalability:** Easy to add new agents
- **Flexibility:** Adapt to different use cases
- **Reliability:** Robust error handling
- **Performance:** Parallel execution where possible

This project demonstrates advanced software engineering, AI integration, and enterprise system design capabilities suitable for a world-class final year project.

